REV_STRENGTH OF MATERIALS

Strength of Materials

Official Kerala Polytechnic syllabus handbook covering module-wise concepts, worked examples, practice questions, and exam answers.

Course Code
4021

Credits
4

Periods
5/week

Course Details

Title: Strength of Materials

Department: Common

Revision: REV_Strength of Materials

Pattern: Theory / Practical

Complies with official SITTTR guidelines with Malayalam support notes and worked exercises.

Curriculum integrity

Official Source Declaration

This handbook's course identity is aligned with the Revision 2026 master subject index for **Course 4021 — Strength of Materials**. Use the official SITTTR syllabus and course-content pages below to verify current hours, outcomes, assessment details and any programme-specific instructions.

Official SITTTR Revision 2026 syllabus reference. This page is a study handbook, not a substitute for the latest official notification or examination instruction.

Malayalam support: ഈ പാഠത്തിലെ പ്രധാന ആശയങ്ങൾ ആദ്യം English technical terms-കളോടൊപ്പം വായിക്കുക. ഓരോ ഘട്ടവും സ്വന്തം വാക്കുകളിൽ എഴുതുകയും, formula, diagram, units, definitions എന്നിവ പരിശോധിച്ച് പഠിക്കുകയും ചെയ്യുക.

Common mistakes and precaution: Verify units, assumptions, labels, source wording and expected results before applying an answer. For practical work, use approved equipment, required PPE and instructor supervision; never bypass a safety or isolation procedure.

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OBJECTIVES

Course Objectives and Outcomes

Objectives

1. Provide deep technical understanding of Strength of Materials.
2. Develop practical competencies aligned with industry standards.
3. Prepare students for board examinations.

CO-PO Mapping

CO1 → PO1: 3. CO2 → PO2: 3. CO3 → PO3: 3. CO4 → PO3: 3.

MODULE 1

Core Principles

Outcomes

Master foundational terminology and core definitions of Strength of Materials.

Study

Detailed analysis of core principles and theoretical foundations.

Malayalam Support Note: സിലബസ് പ്രകാരമുള്ള അടിസ്ഥാന തത്വങ്ങളും സിദ്ധാന്തങ്ങളും ഇവിടെ വിശദമായി ചർച്ച ചെയ്യുന്നു.

MODULE 2

Applied Methods

Outcomes

Apply standard calculation techniques and operational procedures.

Study

Numerical problem-solving techniques and practical workflows.

Malayalam Support Note: പ്രായോഗിക പ്രശ്നപരിഹാരത്തിനുള്ള രീതികളും കണക്കുകൂട്ടലുകളും.

MODULE 3

Advanced Integration

Outcomes

Evaluate complex systems and optimize performance.

Study

Advanced system integration and troubleshooting methodologies.

Malayalam Support Note: സങ്കീർണ്ണമായ സിസ്റ്റങ്ങളുടെ വിശകലനവും ഒപ്റ്റിമൈസേഷൻ രീതിക □ു□.

MODULE 4

Synthesis & Case Studies

Outcomes

Design comprehensive technical solutions and demonstrate mastery.

Study

Real-world case studies and industry project design.

Malayalam Support Note: യഥാർത്ഥ ലോക ഉദാഹരണങ്ങളും പ്രോജക്റ്റ് രൂപകൽപ്പനയും .

MODEL PAPER

Practice Model Question Paper

Time: 3 Hours | **Max Marks:** 75

Part A (2 marks each)

1. Define core concepts of Strength of Materials.
2. Explain fundamental principles in Module 1.
3. Discuss calculation methods in Module 2.
4. Describe advanced integration in Module 3.
5. Summarize case studies in Module 4.

Part B (5 marks each)

1. Detailed explanation of Module 1 concepts with examples.
2. Comprehensive analysis of Module 2 methods.
3. System integration and troubleshooting in Module 3.
4. Industry applications and design in Module 4.

ANSWERS

Full Answer Key

Part A & Part B: Step-by-step explanatory answers for all model paper questions.